# Vidyavardhini's College of Engineering & Technology

Department of Computer Science Engineering (Data Science)

Academic Year: 2026-27

**EXPERIMENT ASSESSMENT**

ACADEMIC YEAR 2026-27

**Course: Wireless and Mobile Communication** **Course code:**

**Year: TE** **SEM: V**

| |
|---|
| Experiment No. 5 |
| **Aim:** To measure the variation of Wi-Fi signal strength with distance from a Wi-Fi router. |
| Name: Meet Mehta |
| Roll Number: 30 |
| Date of Performance: |
| Date of Submission: |

## Evaluation

| Performance Indicator | Max. Marks | Marks Obtained |
|---|---|---|
| Performance | 5 | |
| Understanding | 5 | |
| Journal work and timely submission. | 10 | |
| **Total** | **20** | |

| Performance Indicator | Exceed Expectations (EE) | Meet Expectations (ME) | Below Expectations (BE) |
|---|---|---|---|
| Performance | 5 | 3 | 2 |
| Understanding | 5 | 3 | 2 |
| Journal work and timely submission. | 10 | 8 | 4 |

**Checked by**

**Name of Faculty : Mrs. Sonali Sable**

**Signature : Date :**

**Aim: To measure the variation of Wi-Fi signal strength with distance from a Wi-Fi router.**

## Theory

**1. Wi-Fi Signal Strength**

**Wi-Fi signal strength represents the strength of the wireless signal received by a device from a Wi-Fi router.**

**It is commonly measured using RSSI (Received Signal Strength Indicator) and expressed in dBm.**

**2. RSSI**

**RSSI (Received Signal Strength Indicator) is an indication of the received power level of a wireless signal.**

**RSSI values are generally represented as negative values in dBm.**

**For example:**

- **−40 dBm → stronger signal**
- **−60 dBm → moderate signal**
- **−80 dBm → weaker signal**

**Therefore, as the phone moves farther away from the Wi-Fi router, the received signal strength generally decreases and the RSSI becomes more negative.**

**3. Effect of Distance on Wi-Fi Signal Strength**

**The strength of a Wi-Fi signal generally decreases as the distance between the Wi-Fi router and the receiving device increases.**

**For example, a smartphone placed close to the router may receive a stronger signal than the same smartphone placed farther away.**

**This experiment demonstrates the relationship between distance and received Wi-Fi signal strength.**

**4. Practical Significance**

**Measuring signal strength at different distances helps in understanding:**

- **Wi-Fi coverage area**
- **Effect of distance on wireless communication**
- **Variation of received signal strength**
- **Suitable locations for wireless devices**
- **Limitations of indoor Wi-Fi coverage**

**The actual RSSI values may also be affected by walls, furniture, other wireless networks, obstacles, and device orientation.**

## Procedure:

**Step 1: Set Up the Wi-Fi Router**

**Switch on the Wi-Fi router and keep it at a fixed position.**

**Do not change the router's position during the experiment.**

**Step 2: Open the Wi-Fi Analyzer Application**

**Install and open a suitable Wi-Fi Analyzer application on the smartphone.**

**Step 3: Enable Wi-Fi and Required Services**

**Turn on Wi-Fi on the smartphone.**

**Enable Location services if they are required by the Wi-Fi Analyzer application.**

**Step 4: Connect to the Wi-Fi Network**

**Connect the smartphone to the required Wi-Fi network provided by the router.**

**Verify that the smartphone is connected to the intended network before taking measurements.**

**Step 5: Measure RSSI at 1 m**

**Place the smartphone approximately 1 m away from the Wi-Fi router.**

**Open the Wi-Fi Analyzer application and note the RSSI value in dBm.**

**Step 6: Measure RSSI at Different Distances**

**Move the smartphone away from the router and measure the RSSI at the following distances:**

**2 m, 5 m, 10 m, and 15 m.**

**Record the RSSI value corresponding to each distance.**

**Step 7: Maintain Similar Measurement Conditions**

**Try to keep the orientation and position of the smartphone approximately the same during each measurement.**

**Also avoid unnecessary movement of the router or major changes in the measurement environment.**

**Step 8: Record the Observations**

**Record the measured distance and corresponding RSSI value in the observation table.**

**Step 9: Compare the RSSI Values**

**Compare the RSSI values obtained at different distances.**

**Observe whether the received signal strength decreases as the distance from the router increases.**

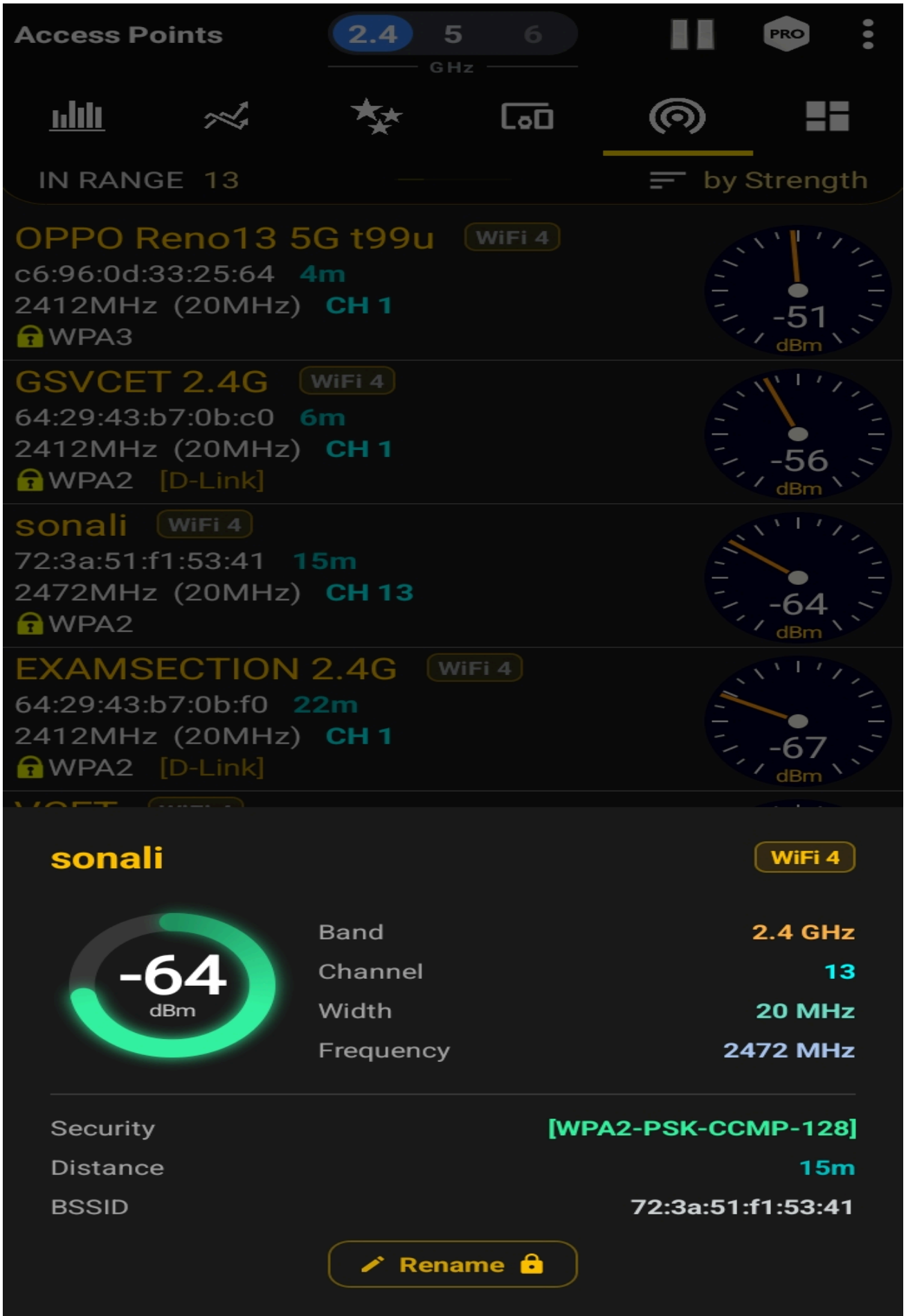
## OUTPUT:

## Observation Table:

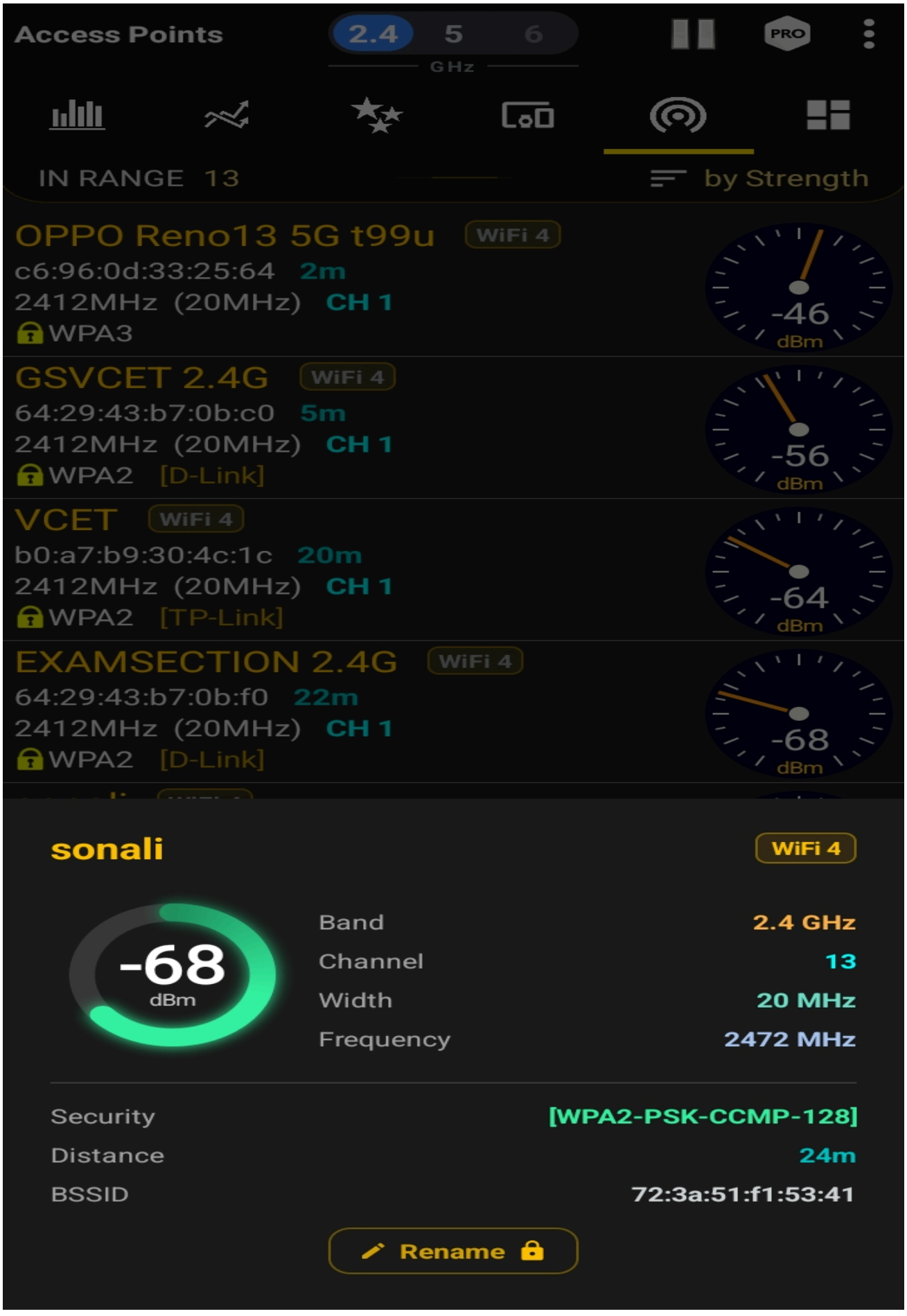
| Sr. No. | Distance from Router (m) | RSSI (dBm) | Signal Strength Observation |
|---|---|---|---|
| 1 | 1 | -25 | Very Strong |
| 2 | 2 | -29 | Strong |
| 3 | 5 | -34 | Strong/ Moderate |
| 4 | 10 | -64 | Moderate |
| 5 | 15 | -68 | Moderate/ Weak |

**Note: Record the RSSI value displayed by the Wi-Fi Analyzer application at each distance. The readings should be taken under approximately the same measurement conditions.**


Access Points
2.4
5
6
GHz
PRO
IN RANGE 8
by Strength
sonali
WiFi 4
72:3a:51:f1:53:41 0m
2472MHz (20MHz) CH 13
WPA2
-25
dBm
OPPO Reno13 5G t99u
WiFi 4
c6:96:0d:33:25:64 11m
2412MHz (20MHz) CH 1
WPA3
-61
dBm
VCET
WiFi 4
b0:a7:b9:30:4c:1c 20m
2412MHz (20MHz) CH 1
WPA2 [TP-Link]
-66
dBm
EXAMSECTION 2.4G
WiFi 4
64:29:43:b7:0b:f0 25m
2412MHz (20MHz) CH 1
WPA2 [D-Link]
-68
dBm
sonali
WiFi 4
-25
dBm
Band
2.4 GHz
Channel
13
Width
20 MHz
Frequency
2472 MHz
Security
[WPA2-PSK-CCMP-128]
Distance
0m
BSSID
72:3a:51:f1:53:41
Rename


Access Points
2.4
5
6
GHz
PRO
IN RANGE 13
by Strength
OPPO Reno13 5G t99u
WiFi 4
c6:96:0d:33:25:64 4m
2412MHz (20MHz) CH 1
WPA3
-51
dBm
GSVCET 2.4G
WiFi 4
64:29:43:b7:0b:c0 6m
2412MHz (20MHz) CH 1
WPA2 [D-Link]
-56
dBm
sonali
WiFi 4
72:3a:51:f1:53:41 15m
2472MHz (20MHz) CH 13
WPA2
-64
dBm
EXAMSECTION 2.4G
WiFi 4
64:29:43:b7:0b:f0 22m
2412MHz (20MHz) CH 1
WPA2 [D-Link]
-67
dBm
sonali
WiFi 4
-64
dBm
Band
2.4 GHz
Channel
13
Width
20 MHz
Frequency
2472 MHz
Security
[WPA2-PSK-CCMP-128]
Distance
15m
BSSID
72:3a:51:f1:53:41
Rename


Access Points
2.4
5
6
GHz
PRO
IN RANGE 13
by Strength
vivo Y39 5G
WiFi 4
f6:e4:73:d8:5e:3e 2m
2412MHz (20MHz) CH 1
WPA2
-47
dBm
OPPO Reno13 5G t99u
WiFi 4
c6:96:0d:33:25:64 10m
2412MHz (20MHz) CH 1
WPA3
-60
dBm
Vivo
WiFi 4
aa:51:4e:f8:54:b7 17m
2437MHz (20MHz) CH 6
WPA2
-65
dBm
sonali
WiFi 4
72:3a:51:f1:53:41 19m
2472MHz (20MHz) CH 13
WPA2
-66
dBm
sonali
WiFi 4
-66
dBm
Band
2.4 GHz
Channel
13
Width
20 MHz
Frequency
2472 MHz
Security
[WPA2-PSK-CCMP-128]
Distance
19m
BSSID
72:3a:51:f1:53:41
Rename


Access Points
2.4
5
6
GHz
PRO
IN RANGE 13
by Strength
OPPO Reno13 5G t99u
WiFi 4
c6:96:0d:33:25:64
2m
2412MHz (20MHz)
CH 1
WPA3
-46
dBm
GSVCET 2.4G
WiFi 4
64:29:43:b7:0b:c0
5m
2412MHz (20MHz)
CH 1
WPA2 [D-Link]
-56
dBm
VCET
WiFi 4
b0:a7:b9:30:4c:1c
20m
2412MHz (20MHz)
CH 1
WPA2 [TP-Link]
-64
dBm
EXAMSECTION 2.4G
WiFi 4
64:29:43:b7:0b:f0
22m
2412MHz (20MHz)
CH 1
WPA2 [D-Link]
-68
dBm
sonali
WiFi 4
-68
dBm
Band
2.4 GHz
Channel
13
Width
20 MHz
Frequency
2472 MHz
Security
[WPA2-PSK-CCMP-128]
Distance
24m
BSSID
72:3a:51:f1:53:41
Rename

**Conclusion :** Thus, the variation of Wi-Fi signal strength with distance was studied using the Wi-Fi Analyzer application. It was observed that RSSI generally becomes more negative as the distance from the router increases. Therefore, Wi-Fi signal strength decreases with increasing distance. Other factors such as walls, obstacles, furniture and smartphone orientation can also affect RSSI.

**VIVA Questions:**

What is RSSI?
RSSI is Received Signal Strength Indicator. It indicates the received power level of a wireless signal.

What happens to Wi-Fi signal strength when the distance from the router increases?
Wi-Fi signal strength generally decreases and RSSI becomes more negative.

Why is RSSI expressed in dBm?
dBm represents signal power relative to 1 milliwatt.

Why should the smartphone orientation be kept approximately the same during measurements?
To maintain similar measurement conditions and avoid changes caused by device orientation.

What factors other than distance affect Wi-Fi signal strength?
Walls, furniture, obstacles, other wireless networks and smartphone orientation.